

SREE BUDDHA COLLEGE OF ENGINEERING, AYATHIL, ELAVUMTHITTA



TOTAL STATION SURVEY REPORT

Year:2022

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Semester: 6

Dept.: CIVIL

CONTOURING

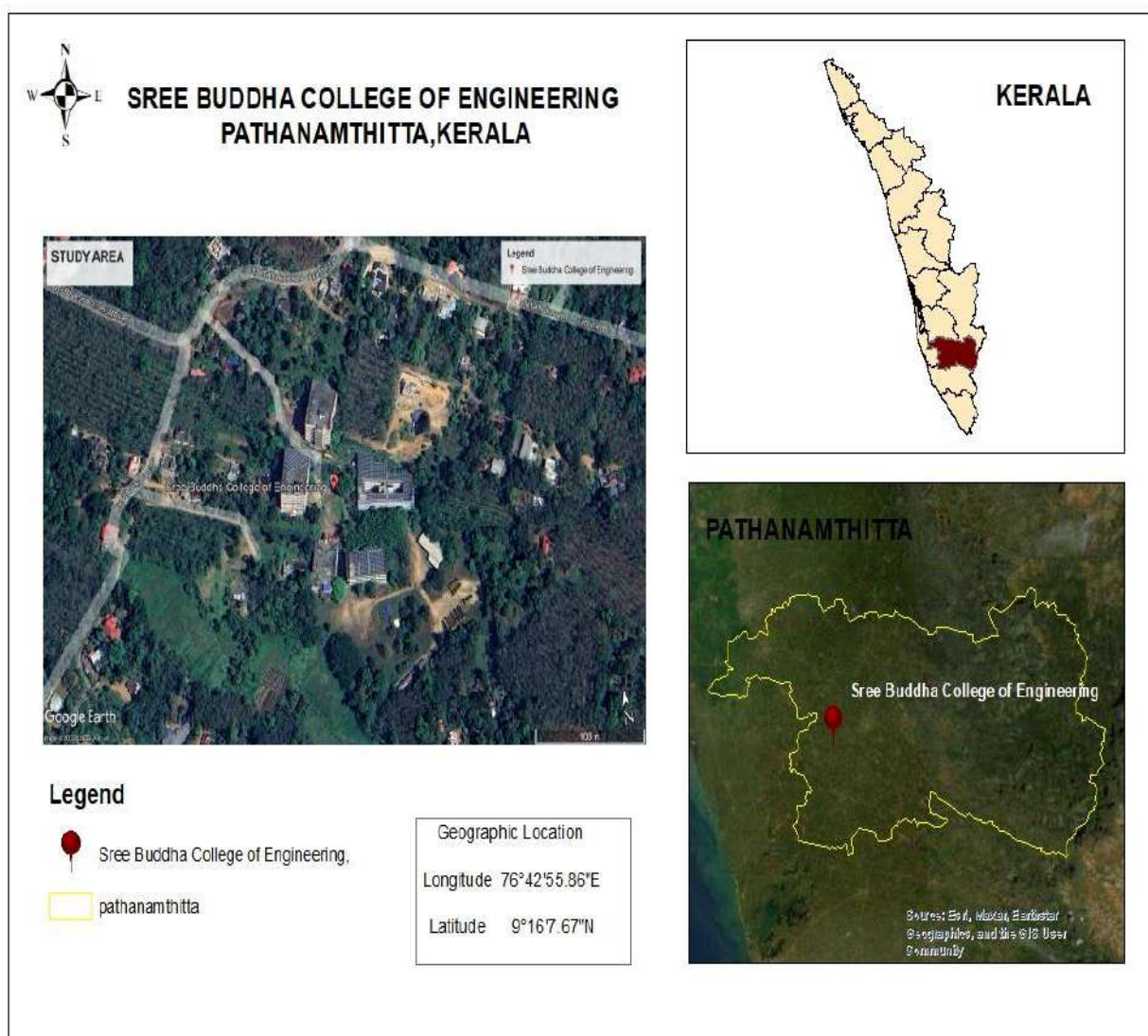
CONTOUR CREATION & MAP CREATION

ALG3

AIM

- Perform a contour survey (by Grid Method using DGPS total station) of the Ground Area of **Sree Buddha College Of Engineering**
- Field exercise on the preparation of contour map for a given terrain using advanced surveying instruments like total station (the survey activity undertaken shall be of at least 5000 sq. m)
- Familiar with field environment and to get practical knowledge of surveying and use of its instruments in field.
- Students learn the use of DGPS total station, they are able to take readings at manual total station, robotic total station, and different level.
- Preparation of contour drawing.

STUDY AREA



CONTOUR

An Imaginary line on the ground surface joining the points of equal elevation is known as contour. The process of tracing contour lines on the surface of the earth is called Contouring.

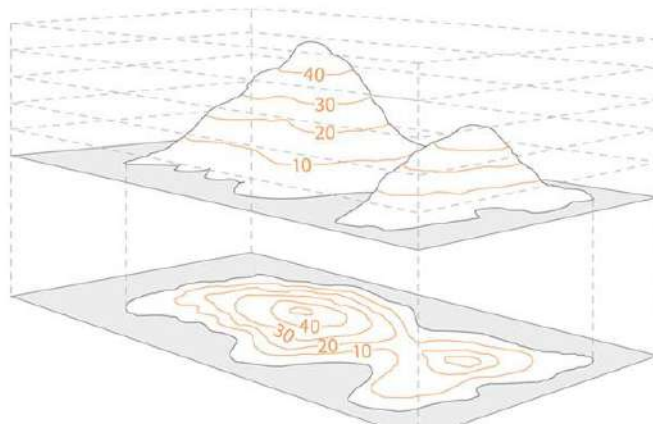


FIG. CONTOUR

CONTOUR INTERVAL

The constant vertical distance between two consecutive contours is called the contour interval

CHARACTERISTICS OF CONTOUR MAPS

The contours maps have the following characteristics:

- All points on a contour line are of the same elevation.
- No two contour lines can meet or cross each other except in the rare case of an overhanging vertical cliff or wall
- Closely spaced contour lines indicate steep slope iv) Widely spaced contour lines indicate gentle slope v) Equally spaced contour lines indicate uniform slope
- Closed contour lines with higher elevation towards the centre indicate hills
- Closed contour lines with reducing levels towards the centre indicate pond or other depression.
- Contour lines of ridge show higher elevation within the loop of the contours. Contour lines cross ridge at right angles.
- Contour lines of valley show reducing elevation within the loop of the contours. Contour lines cross valley at right angles.

- All contour lines must close either within the map boundary or outside.
- Contour lines with U-shape with convexity towards lower ground indicate ridge.
- Contour lines with V-shaped with convexity towards higher ground indicate valley.
- The steepest slope of terrain at any point on a contour is represented along the normal of the contour at that point.
- Contours do not pass through permanent structures such as buildings.
- Contour lines generally do not meet or intersect each other. If contour lines are meeting in some portion, it shows existence of a vertical cliff.
- Contours of different elevations cannot cross each other. If contour lines cross each other, it shows existence of overhanging cliffs or a cave.

FACTORS AFFECTING CONTOUR INTERVAL

Contour interval on a map is decided on the following considerations:

- Scale of the map
- Purpose of the map
- Nature of the ground
- Availability of time and funds

ADVANTAGES OF CONTOUR MAP

1. It is an easy way of measuring and mapping the surface of land.
2. They are useful in topographical engineering, geology, geography, hydrology and civil engineering etc.
3. It is possible to measure distances on sloping land with utmost accuracy by using contour lines.
4. Contour lines are used as a reference in construction and it becomes easier for builders to make precise calculations when building on sloping land or ground because they have something to refer from.
5. Contour lines are being used in mapping the areas for national security, for the purpose of defence.
6. Contour lines are used in setting up boundary limits, to define the boundary of land ownership.

USES OF CONTOUR SURVEYING

- Construction Of Roads And Highways
- Construction Of Dams
- Study Of Relief On Land
- Study Of Vegetation
- Study Of Canals

INSTRUMENT USED: ON FIELD

TOTAL STATION: **TRIMBLE M3 & ROBOTIC TOTAL STATION**

A total station is an electronic/optical instrument used in modern [surveying](#) and [building construction](#) that uses electronic transit theodolite in conjunction with electronic distance meter (EDM). It is also integrated with microprocessor, electronic data collector and storage system.



TOTAL STATION



PRISM POLE



TAP



PRISM

FIELD WORK

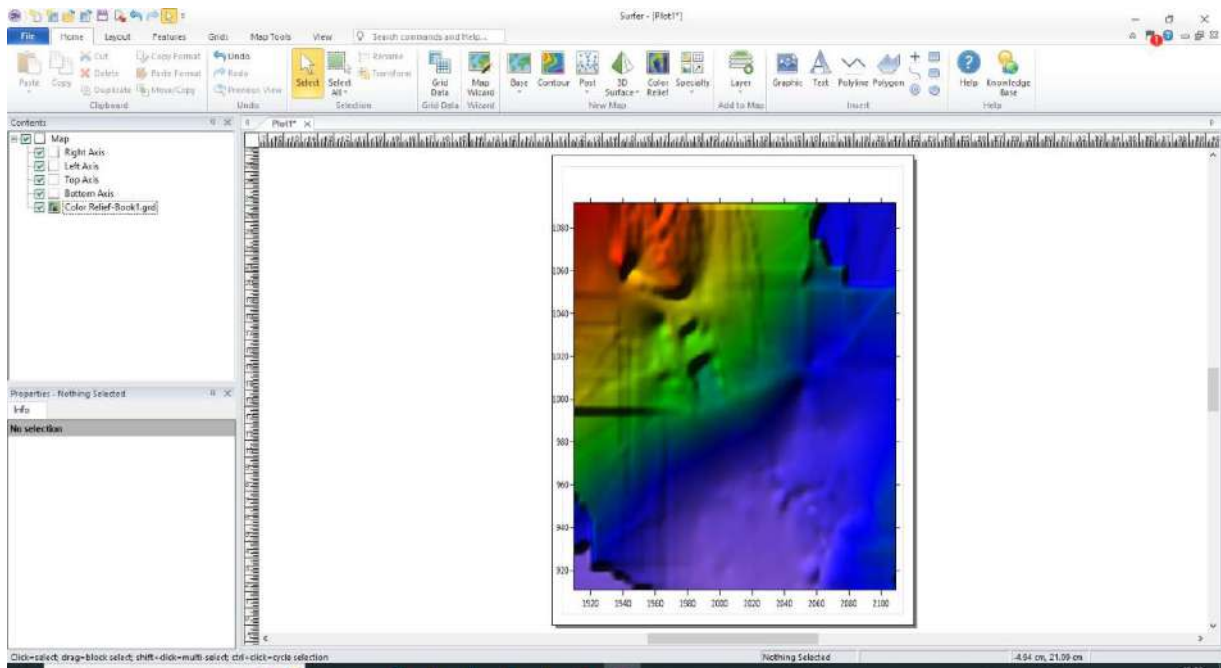


POINT ID	EASTING(X)	NORTHING(y)	ELVATION(Z)	CODE
BACK SIGHT	2006.253	1007.021	100.198	BM
STN 1	2000.000	1000.000	100.000	BS
1	2002.022	1022.357	101.354	rd edg
2	2014.106	1020.842	101.149	rd edg 2
3	2013.611	1018.539	101.198	RC
4	2012.988	1015.862	101.122	RE
5	2007.972	1016.048	101.159	RE
6	2004.289	1016.585	101.244	RE
7	2007.833	1010.792	101.236	RE
8	2007.897	1010.795	101.247	I
9	2006.855	1010.741	101.256	I
10	2006.149	1010.709	101.271	I
11	2005.253	1010.675	101.281	I
12	2003.578	1010.565	101.261	I
13	2001.930	1010.712	101.157	I
14	2001.112	1010.905	101.280	I
15	2000.075	1010.363	101.278	I
16	1999.449	1010.216	101.318	I
17	2000.268	1009.339	100.877	I
18	2001.027	1009.340	100.791	I
19	2002.075	1009.417	100.570	I
20	2003.363	1009.348	100.820	I
21	2004.176	1009.163	100.752	I
22	2005.363	1009.093	100.723	I
23	2006.281	1009.209	100.695	I
24	2007.054	1009.379	100.681	I
25	2007.854	1009.512	100.686	I
26	2008.822	1009.394	100.732	I
27	2009.604	1009.395	100.739	I
28	2010.291	1009.570	100.784	I
29	2010.996	1009.638	100.731	I
30	2011.713	1009.487	100.622	I
31	2011.642	1008.391	100.145	I
32	2010.691	1008.292	100.109	I
33	2009.640	1008.147	100.079	I
34	2008.686	1008.035	100.024	I
35	2007.605	1007.835	100.026	I
36	2005.232	1007.736	100.177	I

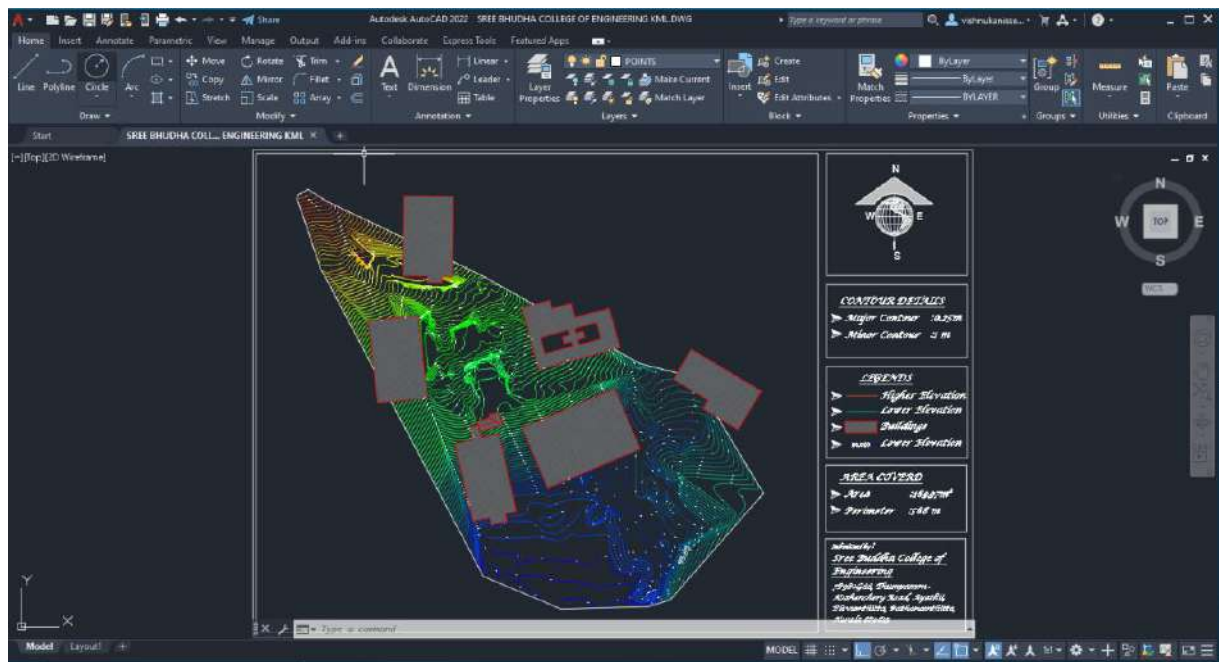
37	2003.980	1007.770	100.131	I
38	2002.801	1008.062	100.203	I
39	2001.893	1007.932	100.291	I
40	2000.671	1007.762	100.188	I
41	2000.019	1007.674	100.175	I

BACKOFFICE WORK

Surfer® is a full-function 2D and 3D mapping, modelling, and analysis software package for scientists and engineers. Surfer's sophisticated interpolation engine quickly transforms XYZ data into publication-quality maps. Virtually every aspect of the map is customizable



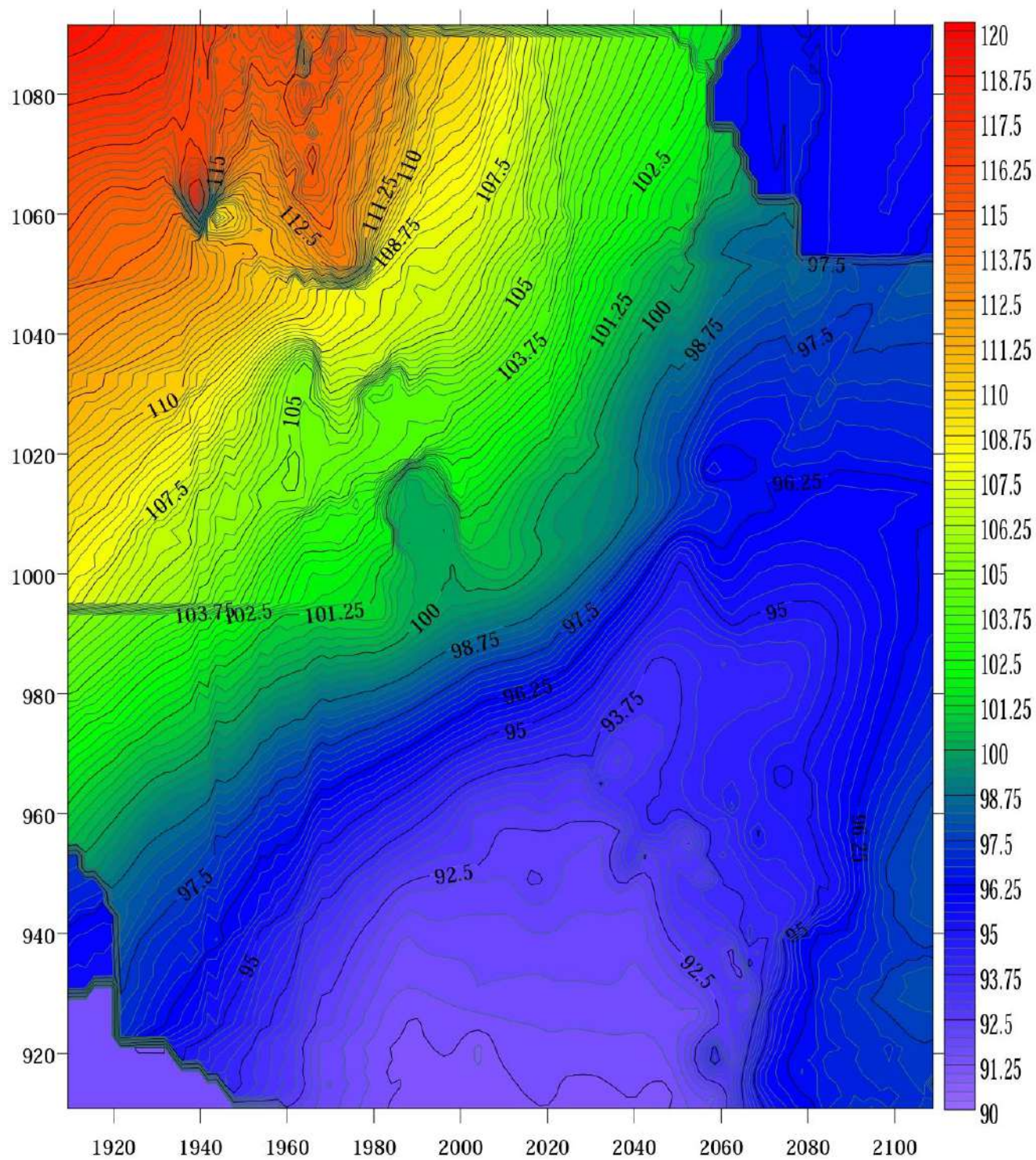
Contour map in Surfer plot



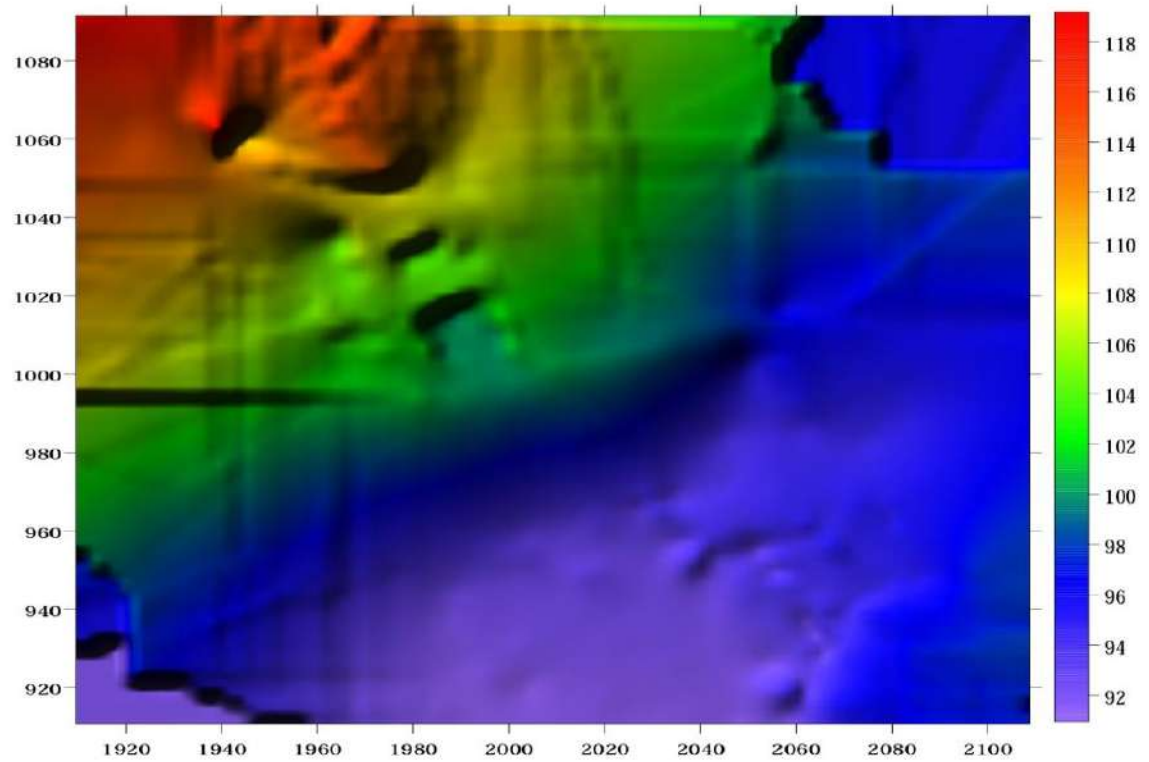
Contour map in civil 3D

S

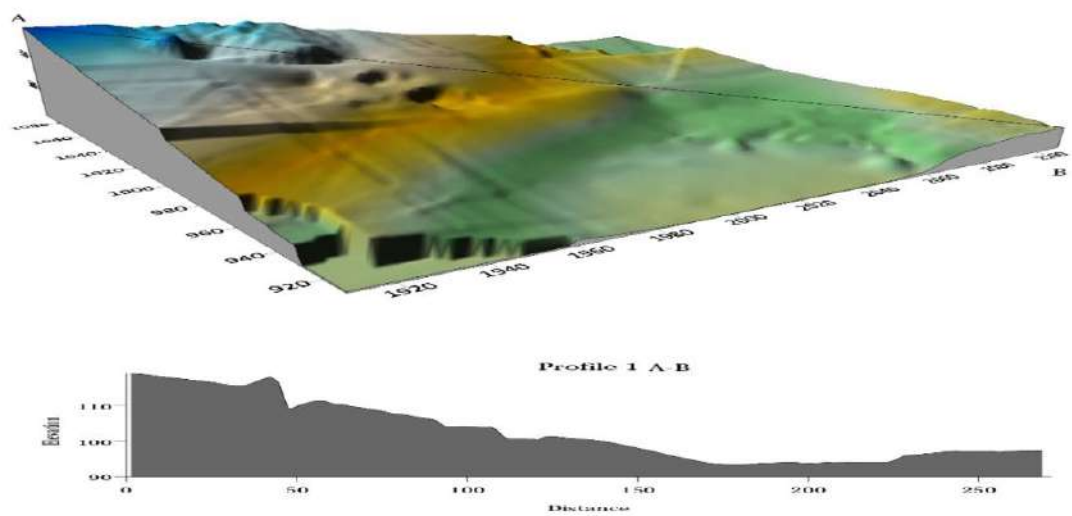
- CONTOUR MAP



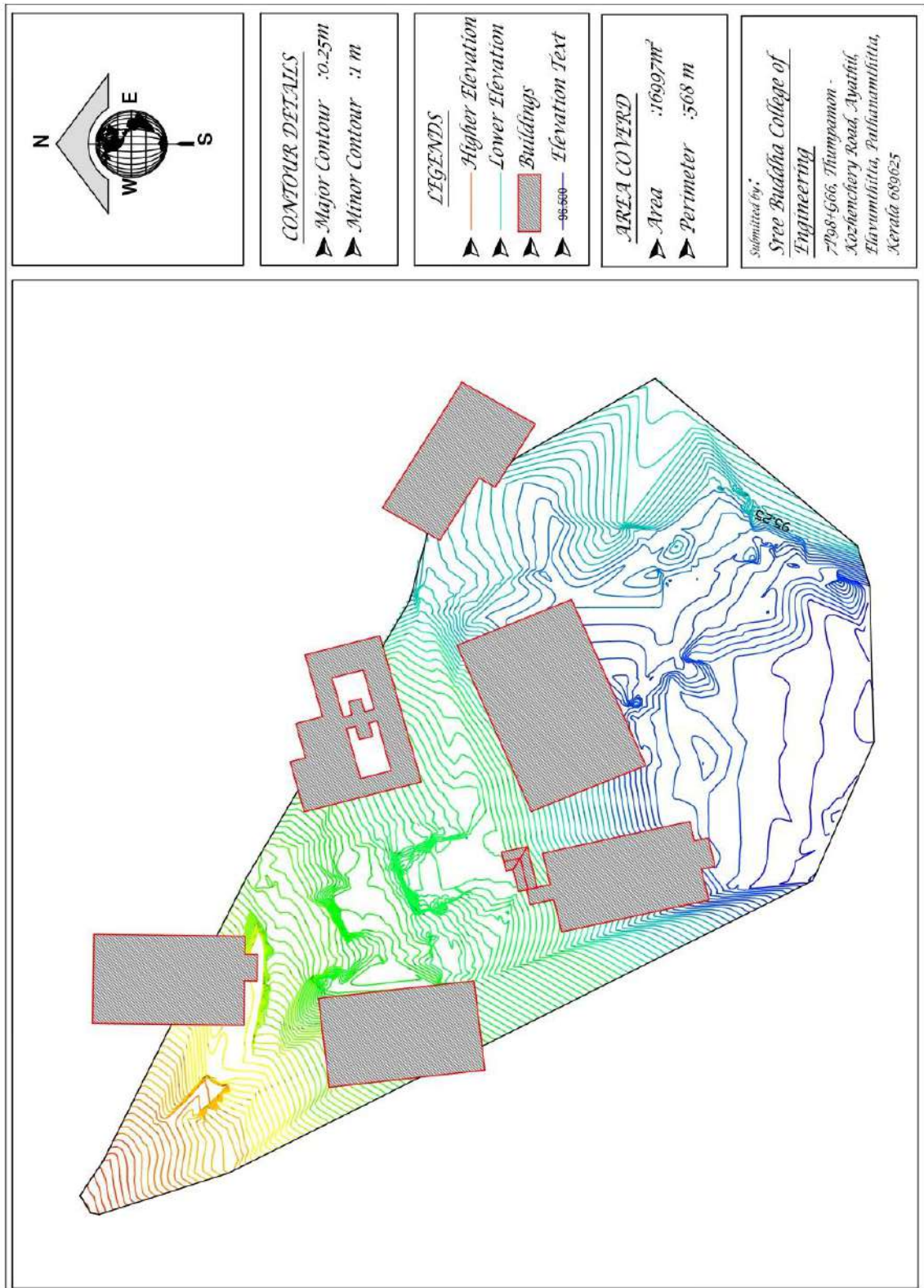
SURFACE MAP



3D SURFACE&CROSS SECTION



AUTOCAD MAP



RESULT:

Preparation of contour construction is completed using Total station.

AUTOCAD MAP

